# PRODUCT USER MANUAL

## For Arctic Ocean BGC Analysis and Forecasting Product

### ARCTIC_ANALYSISFORECAST_BGC_002_004

**Issue: 1.0**

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# RECORD TABLE

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| 1.0 | 2025/05/28 | All | Initial version transferred from common PUM | A Ali, A. Burud, J. Xie, C. Yumruktepe, T. Wakamatsu, A. Melsom, L. Bertino, S. Gao, H. Sumata | L. Bertino |

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# GLOSSARY AND ABBREVIATIONS

| | |
|---|---|
| **MFC** | Monitoring and Forecasting Centre |
| **TAC** | Thematic Assembly Center |
| **ECMWF** | European Centre for Medium Range Weather forecast |
| **GLO** | Global |
| **MFWAM** | Meteo France WAve Model |
| **NRT** | Near Real Time |
| **MY** | Multi Year |
| **PUM** | Product User Manual |
| **QUID** | Quality Information Document |
| **CMEMS** | Copernicus Marine Environment Monitoring Service |
| **NetCDF** | Network Common Data Form |
| **CF** | Climate Forecast (convention for NetCDF) |
| **MDS** | Marine Data Store |
| **FORECAST** | An assessment of the future state of the ocean variables made using numerical models. |
| **HINDCAST** | An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts. |
| **SIMULATION** | An hindcast done without data assimilation. |
| **NRT-ANALYSIS:** | An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis). |
| **BEST-ANALYSIS:** | An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis). |

# DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

# 1) INTRODUCTION

## a) Summary

**ARCTIC_ANALYSISFORECAST_BGC_002_004** is the nominal product of the Arctic Monitoring and Forecasting Center for ocean biogeochemistry, and is composed of 3D, daily mean fields of nitrate, phosphate, silicate, oxygen, chlorophyll, phytoplankton biomass, primary production, pH, dissolved inorganic carbon, and light attenuation.

## b) Polar projections

All products are interpolated onto the same polar stereographic output grid, although at different resolutions. HYCOM uses a staggered Arakawa C grid but all variables are defined at the center of the output grid cell after interpolation.
The same projection is used for all Arctic MFC grids on a spherical Earth, following this definition under the proj4 projection library: (Beware, this is not the WGS84 ellipsoid reference).

projection_stereo:proj4 = "+units=m +proj=stere +a=6378273.0 +b=6378273.0 +lon_0=-45.0 +lat_0=90.0 +lat_ts=90.0 +ellps=sphere" ;

This polar stereographic projection also defines the reference direction of vectors (current and sea ice velocity), which should therefore be rotated if the data is reprojected to a Lat-Long or any other grid.
In ArcGIS and QGIS softwares, the x- and y- coordinates units should be converted from 100 km to meters.

A simple code is provided below :

A MatLab example

```
nx=size(lon,1);
ny=size(lon,2);
radian=pi/180.;
dlon_ip1=(lon(2:nx,:)-lon(1:nx-1,:)).*cos( radian*.5*(lat(2:nx,:)+lat(1:nx-1,:)) );
dlat_ip1=lat(2:nx,:)-lat(1:nx-1,:);
dlon_ip1(nx,:)=dlon_ip1(nx-1,:);
dlat_ip1(nx,:)=dlat_ip1(nx-1,:);
theta_ip1=atan2(dlat_ip1,dlon_ip1); % Angle displacement with constant latitude lines
theta_jp1=theta_ip1+90*radian;
%%rotated Easterly and Northerly components
tmpu=u.*cos(theta_ip1) + v.*cos(theta_jp1);
tmpv=u.*sin(theta_ip1) + v.*sin(theta_jp1);
```

```
A cleaner Python example

import numpy as np
import cartopy.crs as ccrs
import cartopy.feature as cfeature
from cartopy.crs import Projection
from netCDF4 import Dataset

# load data (grid and ocean velocities)
```

```
with Dataset("20250501_dm-metno-MODEL-topaz5_ecosmo-ARC-b20250501-fv02.0.nc") as ds:
    vx = ds["uvo"][0]
    vy = ds["vvo"][0]
    x = ds["x"][:]
    y = ds["y"][:]
    lon = ds["longitude"][:]
    lat = ds["latitude"][:]
spd = np.hypot(vx, vy)
x2d, y2d = np.meshgrid(x, y)

# Define projections
proj_stere = ccrs.Stereographic(central_latitude=90, true_scale_latitude=90,
                                central_longitude=-45,
    globe=ccrs.Globe(semimajor_axis=6378273, semiminor_axis=6378273))
proj_geo = ccrs.PlateCarree()  # lon-lat

# transform velocities from stereographic projection to lon-lat
usi, vsi  = proj_geo.transform_vectors(proj_stere, x2d, y2d, vx, vy)

# transform velocities back from lon-lat to stereographic projection
vx_, vy_  = proj_stere.transform_vectors(proj_geo, lon, lat, u, v)
```

### c) Vertical coordinates

All ARCTIC ocean products – except the sea ice product - are computed on a hybrid vertical coordinate. This is a combination of isopycnal (constant density levels) in the open, stratified ocean, terrain-following coordinates in shallow coastal areas and z-levels in the mixed layer or unstratified deep ocean. See figure below (http://www.hycom.org/hycom/overview).

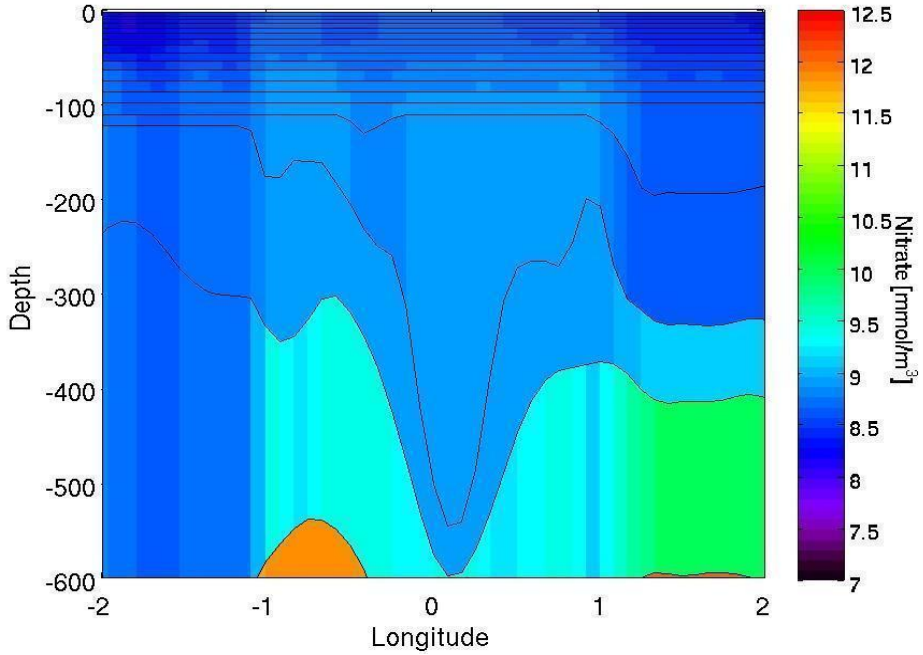



*Figure 6 Vertical section of nitrates and vertical levels. Shading indicates concentrations, thin lines indicate layer interfaces. The coordinates revert from isopycnal to z-levels in the mixed layer.*

The BGC products are interpolated to 40 levels (different from the PHY model): 0, 2, 3, 4, 5, 6, 8, 10, 11, 13, 16, 18, 22, 25, 29, 34, 40, 47, 56, 66, 78, 92, 110, 131, 156, 186, 222, 266, 318, 380, 454, 541, 644, 763, 902, 1062, 2000, 3000, 3500, 4000 m.

The vertical interpolation method from hybrid layers to fixed levels is a cubic spline honouring the following constraints:

- Conservation of the mean value in each layer
- Continuity and derivability of the value at each layer interface
- Vertical derivatives at the uppermost and lowermost model layers.

The cubic splines may overshoot in areas of strong vertical gradients. So after April 2024, the NRT forecasts are using linear interpolation instead. Future product upgrades will use linear interpolation by default.
The model isopycnic layers can become thin at the bottom of the water column (called *outcropping* of the layers) and are removed from the interpolation, resulting in a few bottom grid cells being masked intermittently as land cells. If this issue turns out problematic, for example for downscaling, a turnaround is to apply the masked values constantly to the pixels where the flickering occurs. We will provide a definitive fix to this issue in a future upgrade of the system.

# 2) DESCRIPTION OF THE PRODUCT SPECIFICATION

## a) General Information

1.

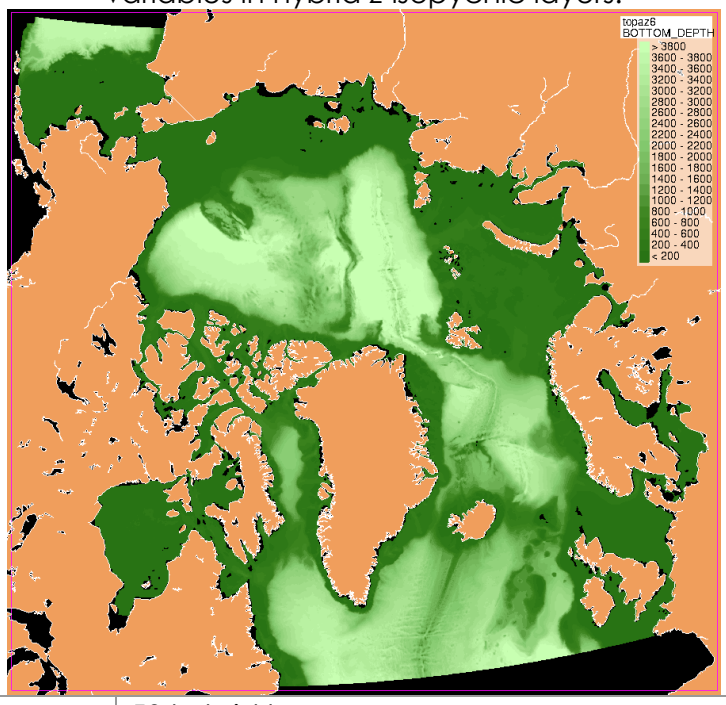
| | |
|---|---|
| **Product Lines** | ARCTIC_ANALYSISFORECAST_BGC_002_004 |
| **Geographical coverage** | North of 63°N and ice-covered North Atlantic Ocean |
| **Variables** | • Nitrate<br>• Phosphate<br>• Silicate<br>• Oxygen<br>• Chlorophyll<br>• Phytoplankton biomass<br>• Primary production<br>• Light attenuation<br>• Model depths<br>• zooplankton<br>• pH<br>• dissolved inorganic carbon<br>• carbon export<br>• Surface CO2 |
| **Product Type** | Near Real Time - Analysis and Forecast |
| **Available time series** | -2Y to 10 Days- forecast |
| **Temporal resolution** | 24hr average and monthly average 3D fields |
| **Update Frequency** | Daily |
| **Target delivery time** | Daily 10-day forecast: daily at 00:30 UTC |
| **Horizontal resolution** | 6.25 km |
| **Number of vertical levels** | 40 levels with a focus on near the surface |
| **Format** | NetCDF4 |

## b) Production System Description

The Arctic MFC biogeochemical ocean system is the TOPAZ5 coupled physical-ecosystem model. The numerical ocean circulation model HYCOM (coupled to a dynamic -thermodynamic sea ice model) is integrated forward in time, forced with atmospheric fields from the ECMWF weather forecast model. The biogeochemical model is ECOSMO-II, coupled to the physics using the FABM software.

ARCTIC (180°W-180°E; 63N – 90°N): polar stereographic grid, 1185x1137 cells

This product is regional with a native grid based on conformal mapping and a very even spatial resolution. It is provided on a standard polar stereographic grid at 6.25 km resolution. All parameters are interpolated from the native Arakawa C grid model to the same output grid. Vertical interpolation is also applied to the 3D variables in hybrid z-isopycnic layers.




| | |
|---|---|
| **Vertical grid** | 50 hybrid layers |
| **System Name** | TOPAZ5-ECOSMO |
| **Model version** | HYCOM v2.2.98 (ocean)<br>CICE 5.1 (sea ice)<br>ECOSMO II (bgc) |
| **Horizontal output Resolution** | 6. 25 km |
| **Vertical output Resolution** | 40 Z levels |
| **Interpolation method** | Horizontally: bilinear, vertically: linear |
| **Coupling** | ESMF and FABM |
| **Sea Ice Model** | CICE v5.1 with 5 ice thickness categories |
| **Bathymetry** | GEBCO14 |
| **Initial conditions** | WOA18 nutrients followed by 20 years of spinup. |

| | |
|---|---|
| **Lateral boundary conditions** | Global HR analysis and forecasting system (PISCES model). |
| **Atmospheric forcings** | Source: ECMWF IFS HRES<br>Frequency: 6-hourly<br>Resolution: 1/10°<br>Other: Completion of solar radiation daily cycle |
| **Atmospheric deposition** | EMEP model nitrate climatology<br>Phosphate based on a N:P ratio. |
| **Tidal constituents** | None |
| **Rivers nutrients** | GlobalNEWS |
| **Data Assimilation scheme** | Empirical projection of satellite surface Chlorophyll-a to profiles according to regions (modified from Ardyna et al. 2013).<br>Direct insertion of resulting plankton profiles. |
| **Assimilated observations** | Global Chlorophyll-a from Ocean Colour satellite |

ECOSMO II(CHL) (Yumruktepe et al, 2022) contains 16 variables: nitrate, ammonium, phosphate, silicate, diatoms, flagellates, cyanobacteria, micro- and meso-zooplankton, particular and dissolved organic matter, opal and oxygen with explicit prognostic chlorophyll for each phytoplankton. Yumruktepe et al. (2022) provide the parameter set that is tuned for the Arctic region. In its recent iteration, ECOSMO II(CHL) received 4 additional state variables (not published), coccolithophores, its chlorophyll-a component, CaCO3 and variable detritus sinking speed respectively. It received a change to its light limitation on growth such that the growth is now directly related to chlorophyll-a concentration (Yumruktepe et al., 2023). ECOSMO II(CHL) parameters were further tuned for the Nordic Seas using chlorophyll-a and oxygen from BGC-Argo buoys. River nutrients were derived from output from the GlobalNEWS model (Seitzinger et al., 2005). At the model lateral boundaries nutrients (nitrate, silicate and phosphate) are nested to the GLO BGC PISCES model through a linearly decreasing effect, otherwise the ECOSMO model variables develop freely. ECOSMO II(CHL) includes an explicit sediment biogeochemistry module which exchanges nutrients, organic matter, oxygen and CaCO3 with the water column. An explicit carbon chemistry module is also included in ECOSMO II(CHL). Horizontally variable annual atmospheric deposition of nitrogen was taken from the EMEP model. Atmospheric deposition of phosphate was resolved using the N:P ratio given in Okin et al. (2011). An ice-algae model (Benkort et al., 2020) has been coupled to the ECOSMO II(CHL) to simulate ice-algae biomass within sea ice. The horizontal advection of the ice-algae related variables is handled like other tracers in CICE. This coupled model facilitates exchanges of algae biomass, nutrients, and detritus between the ice and the surface ocean layer. The presence of ice-algae within the ice affects light penetration and introduces fast sinking organic matter, thereby influencing phytoplankton growth and carbon export in the water column. In the model output, ice-algae biomass is included as part of the phytoplankton biomass in the surface ocean layer, while the ice-algae detritus is added to the export of sinking organic matter.

The ECOSMO II(CHL) model is coupled online to the physical HYCOM model using the FABM coupler. It uses the HYCOM facility for advection of tracers and has the same time-step as the physical model. The interactions between the different components of the model are highlighted in Figure 1. The FABM coupler enables the development of HYCOM and ECOSMO code separately and effectively test the ECOSMO model to be tested in 1D prior to its integration to 3D HYCOM framework.

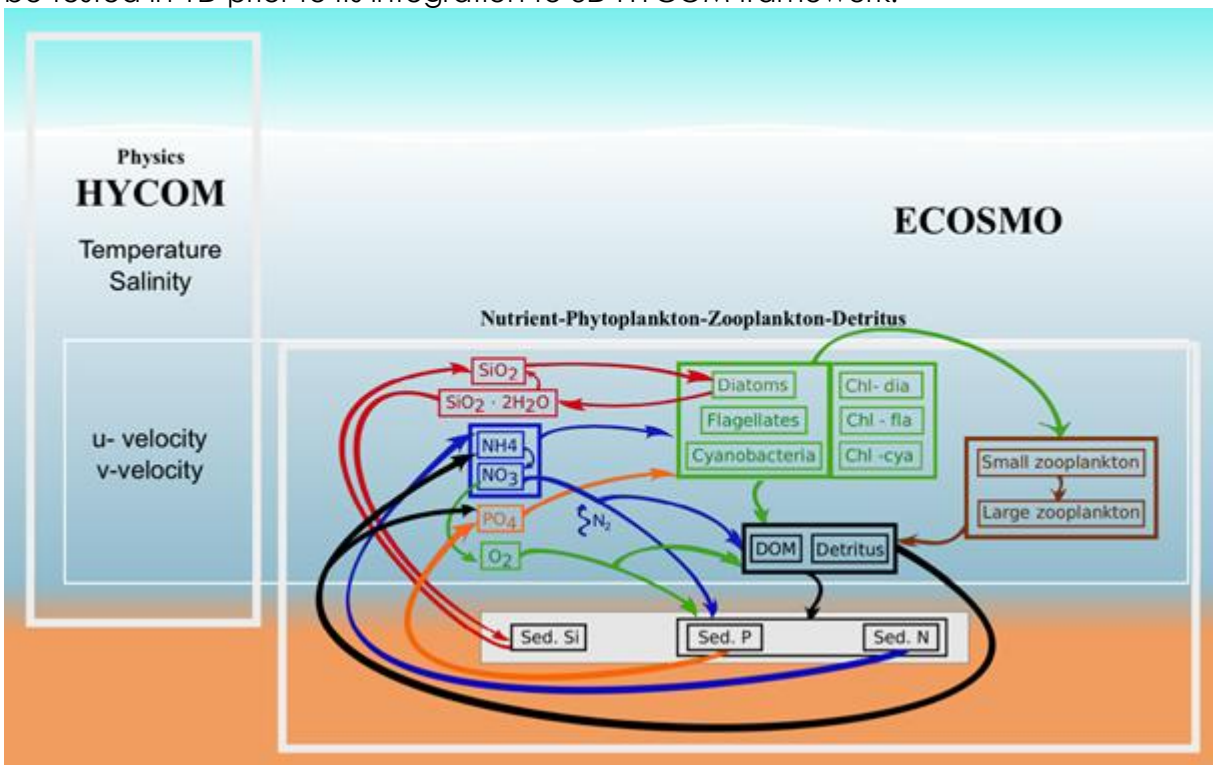



*Figure 1: Overview over the components of ECOSMO and how they interact with each other after Yumruktepe et al. (2022).*

References:

Ardyna, M., Babin, M., Gosselin, M., Devred, E., Bélanger, S., Matsuoka, A., and Tremblay, J.-É.: Parameterization of vertical chlorophyll a in the Arctic Ocean: impact of the subsurface chlorophyll maximum on regional, seasonal, and annual primary production estimates, Biogeosciences, 10, 4383–4404, https://doi.org/10.5194/bg-10-4383-2013, 2013.

Okin, G. S., et al. (2011), Impacts of atmospheric nutrient deposition on marine productivity: Roles of nitrogen, phosphorus, and iron, Global Biogeochem. Cycles, 25, GB2022, doi:10.1029/2010GB003858

Yumruktepe, V. Ç., Samuelsen, A., and Daewel, U.: ECOSMO II(CHL): a marine biogeochemical model for the North Atlantic and the Arctic, Geosci. Model Dev., 15, 3901–3921, https://doi.org/10.5194/gmd-15-3901-2022, 2022.

Yumruktepe, V. Ç., Mousing, E. A., Tjiputra, J., and Samuelsen, A.: An along-track biogeochemical Argo modelling framework, a case study of model improvements for the Nordic Seas, Geosci. Model Dev., 16, 6875–6897, https://doi.org/10.5194/gmd-16-6875-2023, 2023.

Benkort, D., Daewel, U., Heath, M., and Schrum, C.: On the Role of Biogeochemical Coupling Between Sympagic and Pelagic Ecosystem Compartments for Primary and Secondary Production in the Barents Sea, Front. Environ. Sci. 8, 548013, https://doi.org/10.3389/fenvs.2020.548013, 2020.

## c) Dataset details

| | | |
|---|---|---|
| **Product Lines** | ARCTIC_ANALYSISFORECAST_BGC_002_004 | |
| **Geographical coverage** | Arctic | |
| **Datasets:**<br>cmems_mod_arc_bgc_anfc_ecosmo_P1D-m<br>cmems_mod_arc_bgc_anfc_ecosmo_P1M-m | | |
| **Product Type** | Analysis | Forecast |
| **Update frequency** | Daily | Daily |
| **Ensemble** | None | None |
| **Available time series** | -2Y - up to real-time | 10-days forecast |
| **Target delivery time** | | Following day at 00:30 UTC |
| **Temporal resolution (respective datasets)** | Daily mean<br>Monthly mean | |

| **Datasets: cmems_mod_arc_bgc_anfc_ecosmo_P1D-m**<br>**cmems_mod_arc_bgc_anfc_ecosmo_P1M-m**<br>Contain the 3D variables | | | |
|---|---|---|---|
| **Variable** | **Unit** | **Name in file** | **Standard name** |
| Nitrate | mmol m-3 | no3 | mole_concentration_of_nitrate_in_sea_water |
| Phosphate | mmol m-3 | po4 | mole_concentration_of_phosphate_in_sea_water |
| Oxygen | mmol m-3 | o2 | mole_concentration_of_dissolved_molecular_oxygen_in_sea_water |
| Primary production | mg m-3 day-1 | nppv | net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water |
| Chlorophyll | mg m-3 | chl | mass_concentration_of_chlorophyll_a_in_sea_water |
| Silicate | mmol m-3 | si | mole_concentration_of_silicate_in_sea_water |
| Phytoplankton biomass | mmol m-3 | phyc | mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water |
| Light attenuation | m-1 | kd | volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water |
| Model depth | m | model_depth | sea_floor_depth_below_sea_level |

| | | | |
|---|---|---|---|
| Zooplankton biomass | mmol m-3 | zooc | mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water |
| Sinking mole flux of particulate organic matter | mol m-2 d-1 | expc | sinking_mole_flux_of_particulate_organic_matter_expressed_as_carbon_in_sea_water |
| Dissolved inorganic carbon | mole m-3 | dissic | mole_concentration_of_dissolved_inorganic_carbon_in_sea_water |
| Surface partial pressure of carbon dioxide | Pa | spco2 | surface_partial_pressure_of_carbon_dioxide_in_sea_water |
| Sea water ph | 1 | ph | sea_water_ph_reported_on_total_scale |

### d) Additional information on some parameters

None.

### e) Processing information

The TOPAZ5-ECOSMO production follows the same split weekly/daily temporal scheme as the physical forecast (ARCTIC_ANALYSISFORECAST_PHY_002_001). Ocean colour data is assimilated once a week on Thursday using the vertical projection algorithm from Ardyna et al. (2013) with updated parameters for the Norwegian and Barents Sea. The model is run every day in the evening, initialized from a restart file from the same day and produces a 10-day forecast. However on Thursdays, due late arrival of ocean color observations, the model is initialized from the restart file from the previous day on which the ocean color data is assimilated. Therefore the product generated on Thursdays contains a 10-day forecast plus 1 day hindcast. A new monthly mean dataset has been included.

## 2.
## f) Catalogue updating

Assimilation of ocean colour data is performed in a weekly cycle, subsequently right after the physics assimilation is performed on Thursday. 10 days model forecasts are updated on a daily basis, initialized from a restart file from the preceding day, except on Thursday, when the forecast is initialized by the analysis. The assimilation is

performed one day after the satellite observation acquisition, on Thursdays. The ocean colour chlorophyll-a is projected down to 200 meters using the approach of Ardyna et al. (2013) with a further regional parameter modification at the Norwegian and Barents Sea, and the model chlorophyll-a (phytoplankton biomass is modified accordingly) is nudged towards the projected profiles. Consequently, the Thursday model simulation is performed as a one-day hindcast plus 10 days forecast. For the other days of the week, the model simulations are performed such that the first day is using analysed atmospheric forcing fields. A detailed explanation is provided in Table 1.

*Table 1: The weekly production cycle of the ARCTIC_ANALYSISFORECAST_BGC_002_004 product.*

| | | **Valid day** | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Mo | Tu | We | Th | Fr | Sa | Su | Mo | Tu | We | Th | Fr | Sa | Su | Mo | Tu |
| **Bulletin day** | Mo | A* | F | F | F | F | F | F | F | F | F | | | | | | |
| | Tu | | A* | F | F | F | F | F | F | F | F | F | | | | | |
| | We | | | A* | F | F | F | F | F | F | F | F | F | | | | |
| | Th | | | P | A* | F | F | F | F | F | F | F | F | F | | | |
| | Fr | | | | | A* | F | F | F | F | F | F | F | F | F | | |
| | Sa | | | | | | A* | F | F | F | F | F | F | F | F | F | |
| | Su | | | | | | | A* | F | F | F | F | F | F | F | F | F |

*In the table, "Bulletin day" refers to the day when a forecast is produced, and "Valid day" refers to days for which forecasts are available (as daily mean values). Further 'A' indicates simulation with analysed atmospheric forcing fields, 'A*' (on the bulletin day) refers to a mix of analysed atmospheric forcing (up to and including 12 UTC) and forecasts, 'F' corresponds to production using atmospheric forecasts, and 'P' refers to assimilation of the physical state of the model.*

For the daily forecast, the temporal extent is 10 days starting on the bulletin date, except Thursdays when the extent is 13 days starting three days prior to the bulletin date. For the weekly 7-day hindcast provided on Mondays, the temporal extent is 7 days starting on the previous Monday.
The bulletin date is the date at which the production run is scheduled. In the case of the weekly 7-day hindcast, it corresponds to the day after the end of the time period of the data. In the case of the daily forecast, it corresponds to the start of the time period of the data, except for Thursdays where the data start three days earlier than the bulletin date.
The data older than 2 years are automatically removed from the server.. All daily 10-day forecasts are maintained on the dissemination facilities for three months, after which only one forecast per week is retained. All weekly 7-day hindcasts are maintained. Thus, there is always a continuous time series of previous 7-day hindcasts

and the latest forecast available; this is the Best Estimate dataset provided via OPeNDAP/SUBSETTER.

## g) Time and ensemble averaging

The biogeochemical model runs on one member of the TOPAZ5 ensemble without random perturbations.
Forecast and the analysis are 24hr mean fields centered at 12:00 UTC.
The monthly values are centered at the middle of the month.

## h) Product use and impact

The TOPAZ5 system, the physical part of the BGC forecast, is intended to address the broadest range of users in the Nordic Seas and Arctic.
It uses multivariate **data assimilation** with the EnKF, so the users should expect all forecast variables to be accurate to a reasonable degree and consistent with each other: the sea ice edge and the ocean variables affected by sea ice. The TOPAZ5 system does not use an Increment Analysis Update, meaning that the effect of data assimilation appears somewhat abruptly on Tuesday at midnight. The data assimilation software is designed to avoid too sharp data assimilation increments.
The HYCOM model uses hybrid vertical coordinates which limit the noise in **vertical velocities**. This is expected to be advantageous for users running ecosystem models.

# 3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

# 4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

*The ARC MFC products have been set up through different phases of Copernicus Marine and the file and dataset naming conventions have changed in the meantime. The table 2 below shows correspondence between product name, dataset name, and file name.*

*Table 2: File name convention.*

| *Product Name* | *Dataset Name* | *File Name* |
|---|---|---|
| *ARCTIC_ANALYSISFORECAST_BGC_002_004* | *cmems_mod_arc_bgc_anfc_ecosmo_P1D-m* | *20210120_dm-metno-MODEL-topaz5_ecosmo-ARC-b20210111-fv02.0.nc* |
| | *cmems_mod_arc_bgc_anfc_ecosmo_P1M-m* | *202109_mm-metno-MODEL-topaz5_ecosmo-ARC-fv02.0.nc* |

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

## 4.1 FILE SIZE

| DATASET NAME | NAME OF FILE | DIMENSION [GB] |
|---|---|---|
| *cmems_mod_arc_bgc_anfc_ecosmo_P1D-m* | *20210120_dm-metno-MODEL-topaz5_ecosmo-ARC-b20210111-fv02.0.nc* | 0.5 |
| *cmems_mod_arc_bgc_anfc_ecosmo_P1M-m* | *202109_mm-metno-MODEL-topaz5_ecosmo-ARC-fv02.0.nc* | 0.5 |

## 4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

# 5 REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-ARC-QUID-002-004.pdf

# 6 ANNEX

**For dataset cmems_mod_arc_bgc_anfc_ecosmo_P1D-m**

```
Sample file: netcdf \20210120_dm-metno-MODEL-topaz5_ecosmo-ARC-b20210111-
fv02.0 {
dimensions:
   x = 1185 ;
   y = 1137 ;
   depth = 40 ;
   time = UNLIMITED ; // (1 currently)
variables:
   int stereographic ;
         stereographic:grid_mapping_name = "polar_stereographic" ;
         stereographic:latitude_of_projection_origin = 90. ;
         stereographic:longitude_of_projection_origin = -45. ;
         stereographic:scale_factor_at_projection_origin = 1. ;
         stereographic:straight_vertical_longitude_from_pole = -45. ;
         stereographic:earth_radius = 6378273. ;
         stereographic:proj4 = "+proj=stere +lon_0=-45 +lat_0=90 +k=1 +R=6378273
+no_defs" ;
         stereographic:false_easting = 0. ;
         stereographic:false_northing = 0. ;
   double time(time) ;
         time:units = "seconds since 1970-1-1T00:00:00Z" ;
         time:long_name = "forecast time" ;
   float x(x) ;
         x:axis = "X" ;
         x:standard_name = "projection_x_coordinate" ;
         x:units = "100  km" ;
   float longitude(y, x) ;
         longitude:standard_name = "longitude" ;
         longitude:units = "degrees_east" ;
   float y(y) ;
         y:standard_name = "projection_y_coordinate" ;
         y:axis = "Y" ;
         y:units = "100  km" ;
   float latitude(y, x) ;
         latitude:standard_name = "latitude" ;
         latitude:units = "degrees_north" ;
   float depth(depth) ;
         depth:long_name = "depth" ;
         depth:units = "m" ;
         depth:standard_name = "depth" ;
         depth:positive = "down" ;
         depth:axis = "Z" ;
   float model_depth(y, x) ;
         model_depth:_FillValue = -1.e+14f ;
```

```
            model_depth:missing_value = -1.e+14f ;
            model_depth:units = "meter" ;
            model_depth:standard_name = "sea_floor_depth_below_sea_level" ;
            model_depth:grid_mapping = "stereographic" ;
            model_depth:coordinates = "longitude latitude" ;
            model_depth:cell_methods = "area: mean" ;
    float nppv(time, depth, y, x) ;
            nppv:_FillValue = -1.e+14f ;
            nppv:missing_value = -1.e+14f ;
            nppv:units = "mg m-3 day-1" ;
            nppv:standard_name =
"net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_
water" ;
            nppv:grid_mapping = "stereographic" ;
            nppv:coordinates = "longitude latitude" ;
            nppv:cell_methods = "area: mean" ;
    float chl(time, depth, y, x) ;
            chl:_FillValue = -1.e+14f ;
            chl:missing_value = -1.e+14f ;
            chl:units = "mg m-3" ;
            chl:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
            chl:grid_mapping = "stereographic" ;
            chl:coordinates = "longitude latitude" ;
            chl:cell_methods = "area: mean" ;
    float kd(time, depth, y, x) ;
            kd:_FillValue = -1.e+14f ;
            kd:missing_value = -1.e+14f ;
            kd:units = "m-1" ;
            kd:standard_name =
"volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water" ;
            kd:grid_mapping = "stereographic" ;
            kd:coordinates = "longitude latitude" ;
            kd:cell_methods = "area: mean" ;
    float no3(time, depth, y, x) ;
            no3:_FillValue = -1.e+14f ;
            no3:missing_value = -1.e+14f ;
            no3:units = "mmol m-3" ;
            no3:standard_name = "mole_concentration_of_nitrate_in_sea_water" ;
            no3:grid_mapping = "stereographic" ;
            no3:coordinates = "longitude latitude" ;
            no3:cell_methods = "area: mean" ;
    float po4(time, depth, y, x) ;
            po4:_FillValue = -1.e+14f ;
            po4:missing_value = -1.e+14f ;
            po4:units = "mmol m-3" ;
            po4:standard_name = "mole_concentration_of_phosphate_in_sea_water" ;
            po4:grid_mapping = "stereographic" ;
            po4:coordinates = "longitude latitude" ;
            po4:cell_methods = "area: mean" ;
```

```
    float phyc(time, depth, y, x) ;
            phyc:_FillValue = -1.e+14f ;
            phyc:missing_value = -1.e+14f ;
            phyc:units = "mmol m-3" ;
            phyc:standard_name =
"mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water" ;
            phyc:grid_mapping = "stereographic" ;
            phyc:coordinates = "longitude latitude" ;
            phyc:cell_methods = "area: mean" ;
    float zooc(time, depth, y, x) ;
            zooc:_FillValue = -1.e+14f ;
            zooc:missing_value = -1.e+14f ;
            zooc:units = "mmol m-3" ;
            zooc:standard_name =
"mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water" ;
            zooc:grid_mapping = "stereographic" ;
            zooc:coordinates = "longitude latitude" ;
            zooc:cell_methods = "area: mean" ;
    float o2(time, depth, y, x) ;
            o2:_FillValue = -1.e+14f ;
            o2:missing_value = -1.e+14f ;
            o2:units = "mmol m-3" ;
            o2:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
            o2:grid_mapping = "stereographic" ;
            o2:coordinates = "longitude latitude" ;
            o2:cell_methods = "area: mean" ;
    float si(time, depth, y, x) ;
            si:_FillValue = -1.e+14f ;
            si:missing_value = -1.e+14f ;
            si:units = "mmol m-3" ;
            si:standard_name = "mole_concentration_of_silicate_in_sea_water" ;
            si:grid_mapping = "stereographic" ;
            si:coordinates = "longitude latitude" ;
            si:cell_methods = "area: mean" ;
    float expc(time, depth, y, x) ;
            expc:_FillValue = -1.e+14f ;
            expc:missing_value = -1.e+14f ;
            expc:units = "mol m-2 d-1" ;
            expc:standard_name =
"sinking_mole_flux_of_particulate_organic_matter_expressed_as_carbon_in_sea_water
" ;
            expc:grid_mapping = "stereographic" ;
            expc:coordinates = "longitude latitude" ;
            expc:cell_methods = "area: mean" ;
    float ph(time, depth, y, x) ;
            ph:_FillValue = -1.e+14f ;
            ph:missing_value = -1.e+14f ;
            ph:units = "1" ;
```

```
            ph:standard_name = "sea_water_ph_reported_on_total_scale" ;
            ph:grid_mapping = "stereographic" ;
            ph:coordinates = "longitude latitude" ;
            ph:cell_methods = "area: mean" ;
    float dissic(time, depth, y, x) ;
            dissic:_FillValue = -1.e+14f ;
            dissic:missing_value = -1.e+14f ;
            dissic:units = "mole m-3" ;
            dissic:standard_name =
"mole_concentration_of_dissolved_inorganic_carbon_in_sea_water" ;
            dissic:grid_mapping = "stereographic" ;
            dissic:coordinates = "longitude latitude" ;
            dissic:cell_methods = "area: mean" ;
    float spco2(time, depth, y, x) ;
            spco2:_FillValue = -1.e+14f ;
            spco2:missing_value = -1.e+14f ;
            spco2:units = "Pa" ;
            spco2:standard_name =
"surface_partial_pressure_of_carbon_dioxide_in_sea_water" ;
            spco2:grid_mapping = "stereographic" ;
            spco2:coordinates = "longitude latitude" ;
            spco2:cell_methods = "area: mean" ;

// global attributes:
            ;
             :institution = "Met Norway, Henrik Mohns plass 1, 0313 Oslo,  Norway" ;

            :source = "NERSC-HYCOM model fields" ;

              :references = "https://marine.copernicus.eu/" ;
            :
            :conventions = "CF-1.4" ;
             :contact = "<servicedesk.cmems@mercator-ocean.eu>" ;
             :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
            :field_date = "2021-01-20T12:00:00Z" ;
            :bulletin_date = "2021-01-11" ;
            :title = "Arctic Ocean Biogeochemistry Analysis and Forecast, 6.25 km daily
mean" ;
            :bulletin_type = "Forecast" ;
            :Forecast_range = "10 days" ;
}
```